PROJECT DESIGN PHASE

Solution Architecture

Software Installation Request (ServiceNow)

Date	20 October 2026
Team ID	
Project Name	Software Installation Request (ServiceNow)
Maximum Marks	4 Marks

1. Solution Architecture

The Software Installation Request solution is a ServiceNow Service Catalog process that lets an employee request installation of company-approved software. The request captures the software name, version, business reason, and urgency. Workflow Studio requests manager approval; when approved, it creates a Catalog Task for the Software Support Team. The flow retrieves the submitted catalog variables, then monitors the task until it is marked Closed Complete or Pending, which drives the corresponding requester notification. A separate Requested Item Business Rule is intended to create an Incident when the Requested Item enters On Hold (state value 3).

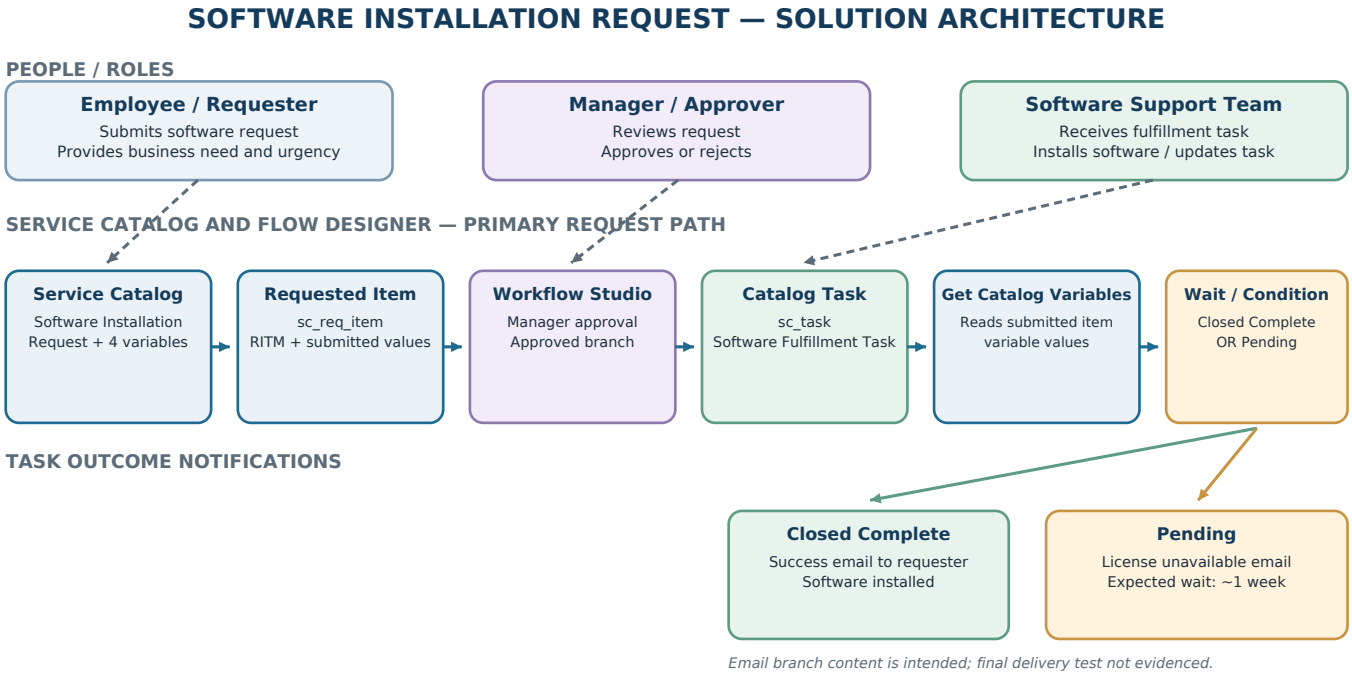


Figure 1. Conceptual architecture and process flow based on the configured catalog item, Workflow Studio flow, Catalog Task, and separate Incident Business Rule.

2. Architecture Components and Flow

Component / Actor	Responsibilities / Activities	Technology / Layer
Employee / Requester	Submits the catalog request and provides software name, requested version, business reason, and urgency. Receives the outcome emails.	Service Catalog UI; Requested for user record
Service Catalog Item	Presents the Software Installation Request form. The item is active, in the Software category, and belongs to the Service Catalog.	Catalog Item [sc_cat_item]; four catalog variables
Requested Item / Request	Stores the submitted request context and links it to the generated fulfillment task. Observed example: RITM0010003 under REQ0010003.	Requested Item [sc_req_item]; Request [sc_request]
Manager / Approver	Reviews the request through the approval action. Configured rule: Anyone approves; approver is the requester's manager.	Ask For Approval; Approval field and Approval history
Workflow Studio / Flow Designer	Coordinates approval, approved-branch task creation, catalog-variable retrieval, task-state waiting, and outcome email actions.	Software Request Workflow; Service Catalog trigger; Flow Designer actions
Software Support Team	Receives the Software Fulfillment Task and updates its state after installation or when a license is unavailable.	Catalog Task [sc_task]; assignment group Software Support Team
Incident Business Rule	Separately checks the Requested Item state in script and inserts an Incident when state equals 3 (commented as On Hold).	Create Incident on License Issue; rule on sc_req_item; Incident [incident]
ServiceNow data layer	Maintains catalog item definitions, request records, approvals, fulfillment tasks, user references, and Incident records.	ServiceNow platform tables and reference fields

3. Main Solution Functions

• Employee submits a Software Installation Request with four catalog variables: software name, version, business reason, and urgency.

• Workflow Studio asks the requester's manager for approval and evaluates the approved branch.

• On approval, the flow creates a Catalog Task linked to the Requested Item, with short description Software Fulfillment Task and assignment group Software Support Team.

• The flow retrieves catalog variables using Get Catalog Variables and waits for the Catalog Task to become Closed Complete or Pending.

• Closed Complete is intended to trigger a successful-installation email; Pending is the project's chosen signal for license unavailability and triggers an email explaining an expected delay of approximately one week.

• A separate Business Rule is intended to create an Incident when the Requested Item state is On Hold (3).

4. System Data Flow

Primary request path: Employee → Service Catalog item → Requested Item (sc_req_item) → manager approval in Workflow Studio → Catalog Task (sc_task) assigned to Software Support Team → Get Catalog Variables → Wait For Condition → outcome email to Requested for user.

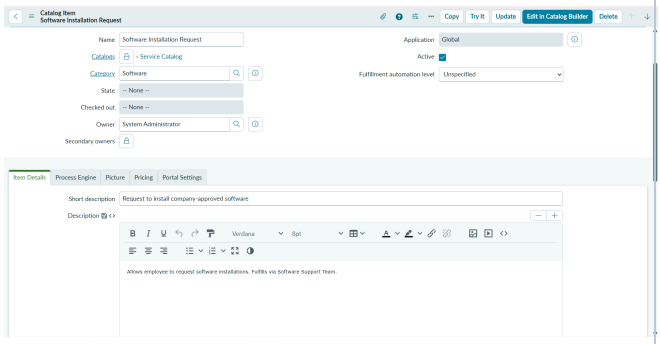
Outcome handling: Catalog Task state Closed Complete routes to the installation-completed notification. Catalog Task state Pending routes to the license-unavailable notification, which states that a new license must be obtained and installation is expected in approximately one week.

Separate incident path: Requested Item state value 3 (commented as On Hold) → Create Incident on License Issue Business Rule → new Incident record.

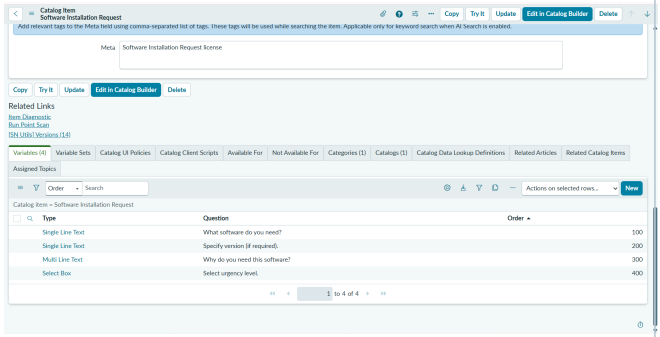
Implementation and verification notes
• The project uses Workflow Studio / Flow Designer; Legacy Workflow is not part of the design.
• Pending is a project-specific Catalog Task signal for license shortage. It is not interchangeable with On Hold on the Requested Item.
• The flow screenshot shows Get Catalog Variables in the sequence, but the action inputs and selected output pills were not visible in the supplied evidence.
• The Business Rule screenshot shows the Filter Conditions area blank. Its trigger condition and the script's group/caller references require validation.
• Personalized email wording and final email delivery for both branches were not confirmed by end-to-end evidence.

5. Configuration Evidence — Catalog and Request Records

The screenshots below are from the supplied ServiceNow project evidence. They show the configured catalog item, its four variables, and an example submitted Requested Item record.

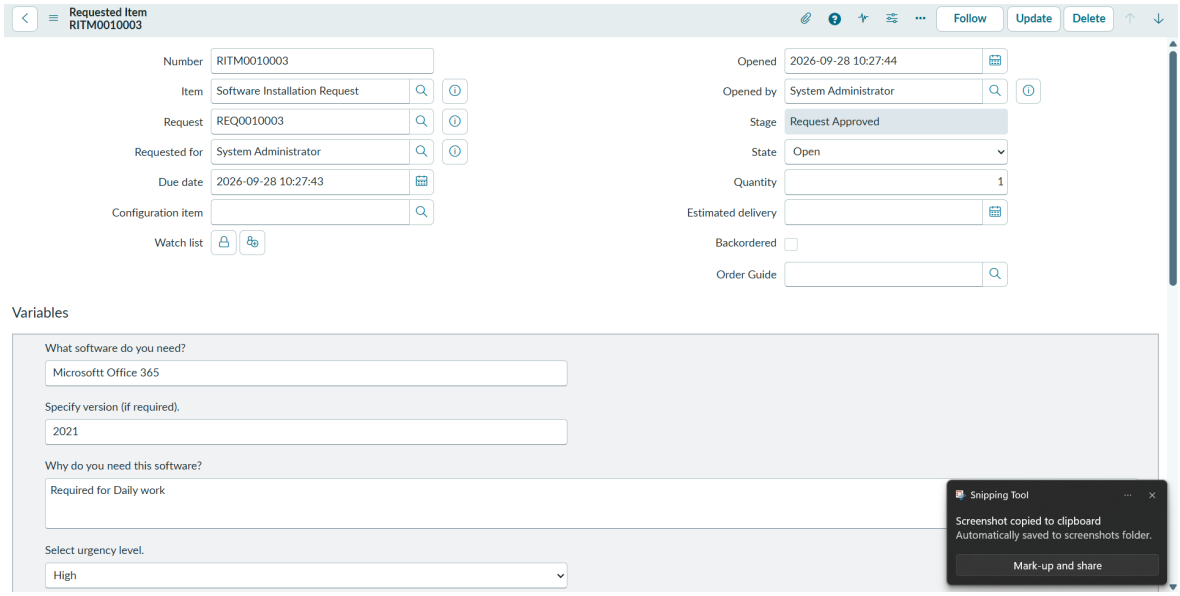


Active Software Installation Request item, Software category, Service Catalog, and company-approved software description.



Four variables in order: software name, version, business reason, and urgency.

Figure 4. Submitted Requested Item — RITM0010003



Observed record details: Item = Software Installation Request; Request = REQ0010003; Requested for = System Administrator; Stage = Request Approved; State = Open. Submitted values shown: Microsoft Office 365, version 2021, business reason "Required for Daily work," and urgency High.

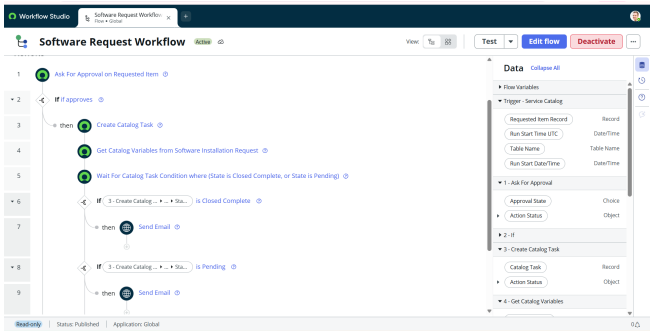
Catalog Variable Definition Summary

Question shown to requester	Type	Internal name evidence	Order
What software do you need?	Single Line Text	Software_Name (confirmed)	100
Specify version (if required).	Single Line Text	Not visible in supplied definition screenshot	200
Why do you need this software?	Multi Line Text	Not visible in supplied definition screenshot	300
Select urgency level.	Select Box	Not visible in supplied definition screenshot	400

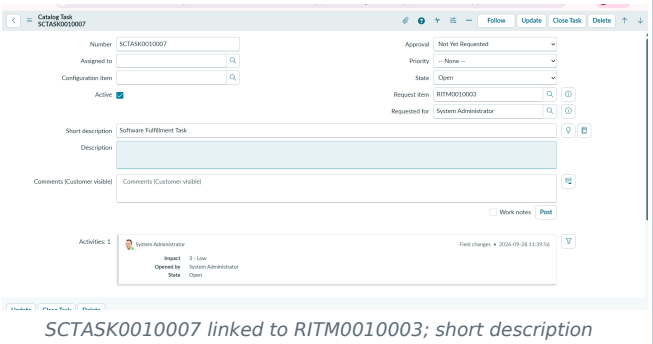
Only the internal variable name Software_Name was visible in the supplied variable-definition screenshot; the remaining internal names are intentionally not inferred.

6. Configuration Evidence — Automation and Deployment

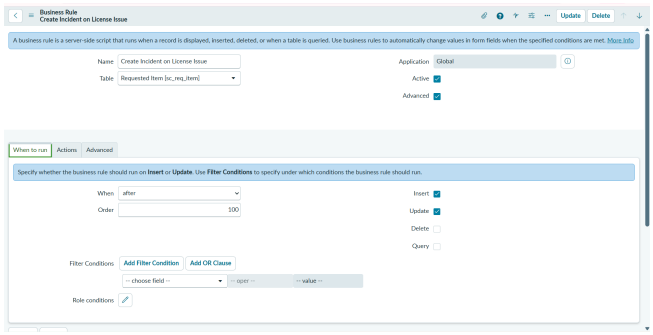
These screenshots document the Workflow Studio sequence, the generated fulfillment task, the separate Business Rule configuration, and the update-set commit result.



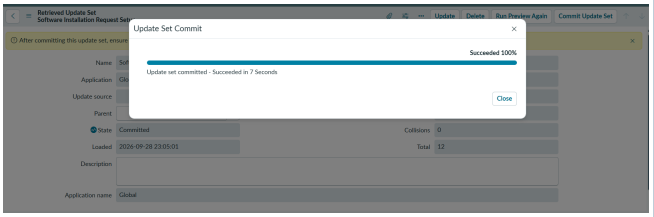
Published flow sequence: approval, approved branch, Create Catalog Task, Get Catalog Variables, wait for Closed Complete or Pending, and two Send Email branches.



SCTASK0010007 linked to RITM0010003; short description Software Fulfillment Task; Requested for System Administrator; state Open.



Create Incident on License Issue on Requested Item [sc_req_item], active and advanced, after insert/update. The displayed filter condition is blank and requires verification.



Software Installation Request Setup commit succeeded 100%; state Committed; 0 collisions; 12 total updates.

7. Key Design Distinctions and Verification Status

Record separation: The workflow monitors Catalog Task [sc_task], while the Business Rule is attached to Requested Item [sc_req_item]. These are different records and their state values must not be conflated.

License shortage: The workflow uses Catalog Task = Pending as the project-specific shortage signal because this was the available task state chosen for the project. The Business Rule script separately checks Requested Item state == 3, commented as On Hold.

Assignment groups: The Catalog Task uses Software Support Team. The Business Rule script sets its Incident assignment group by display value to Software Support. The two names differ and should not be assumed to reference the same group.

Observed validation: A Workflow Studio test run showed the approval action completed, the approval condition evaluated True, and Create Catalog Task completed. The update set commit succeeded with 12 updates and zero collisions. The supplied evidence does not establish that both task-state email branches or Incident creation were end-to-end tested.

Conclusion: The architecture connects employee software requests, manager approval, support fulfillment, requester notifications, and separate license-issue Incident automation using ServiceNow-native components. The committed update set and observed request/task records provide implementation evidence; outstanding items are verification of the Business Rule condition and references, Get Catalog Variables outputs, and the two real email outcome tests.